

TERRALINK

BRAINSTORMING:

TerraLink was developed through brainstorming **different communication technologies and network approaches** to create a reliable, decentralized emergency communication system.



IDEA PRIORITIZATION MATRIX

S.No	Idea / Feature	Impact	Feasibility	Priority	Decision
1	LoRa-based Communication	High	High	Very High	Implement
2	GPS-based SOS	High	High	Very High	Implement
3	Multi-hop Communication	Very High	Medium	Very High	Implement
4	Dynamic Relay Nodes	Very High	Medium	Very High	Core Innovation
5	Dedicated Rescue Relay	High	High	High	Implement
6	Controlled Packet Forwarding	High	High	High	Implement
7	Battery-aware Relay Selection	High	Medium	High	Develop
8	Dynamic Route Recovery	Very High	Medium/Low	Medium	Future Development
9	Solar-powered Relay	Medium	Medium	Medium	Future Enhancement
10	Intelligent Energy Routing	Very High	Low/Medium	Medium	Future Research
11	Drone-based Relay	High	Low	Low	Long-term Idea

Final Selection:

LoRa-based multi-hop mesh communication was prioritized because it provides the best balance of:

“Long range + Low power + Low infrastructure dependency + Multi-hop reliability”

For TerraLink, this led to the development of relay nodes, SOS messaging, GPS location sharing, and self-healing communication.

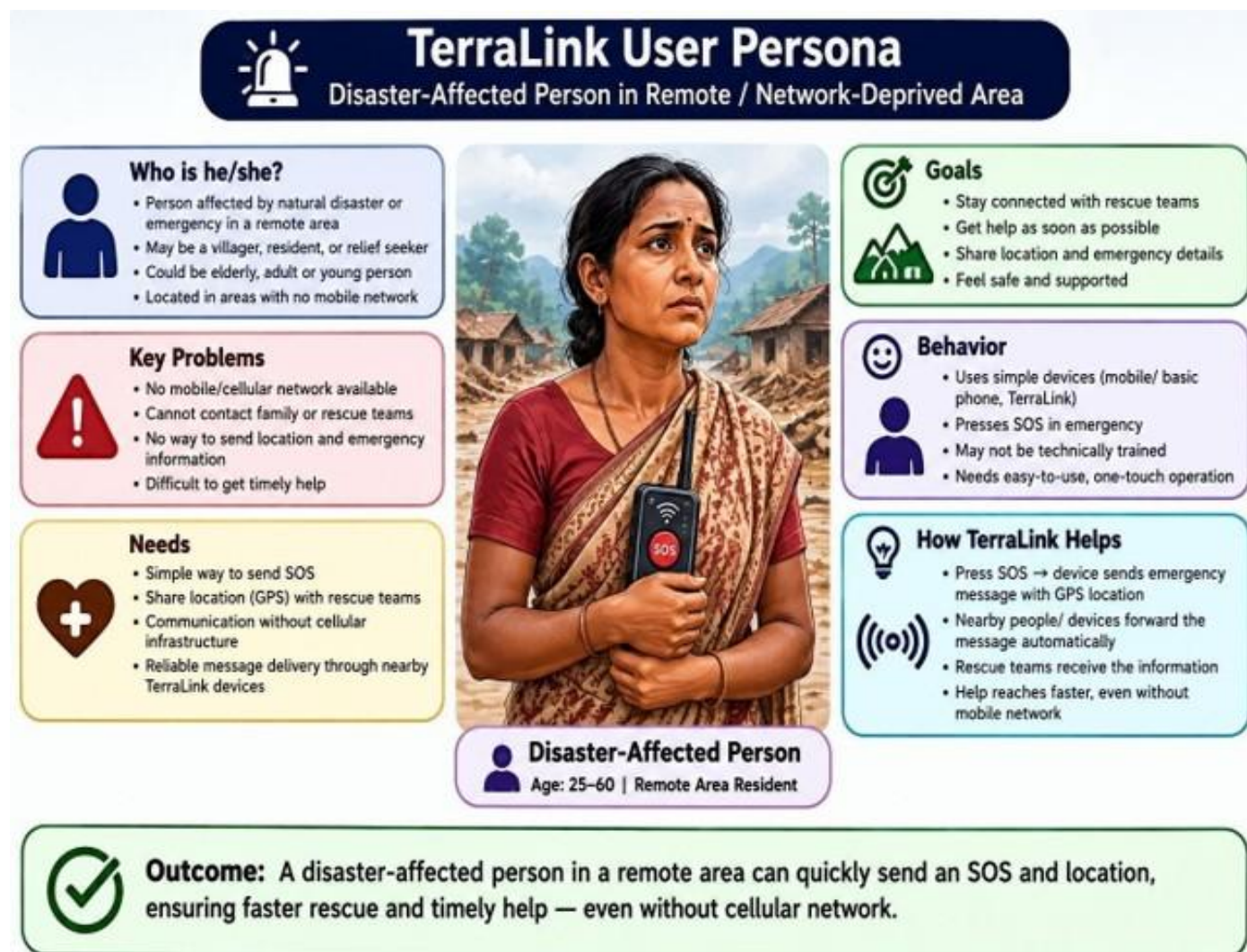
EMPATHY MAP:

SAYS • "What if I lose mobile signal in there?" • "How do I even contact anyone from this spot?" • "I need my exact location to reach the rescue team, not just 'somewhere near the ridge.'" • "I don't want to carry another complicated gadget on top of everything else." • "Is there anyone else out here who could pass this along for me?"	THINKS • What if I become completely unreachable out here? • Will anyone actually respond in time if I do send an SOS? • Do I have enough battery left for an emergency later? • Is my GPS position accurate right now, or still acquiring? • Did my SOS actually reach someone, or did it just vanish? • Is there even another TerraLink user anywhere near me to forward this?
DOES • Carries a phone and basic emergency/trekking equipment. • Checks cellular network availability out of habit, expecting none. • Uses GPS/offline maps for daytime navigation. • Moves toward higher or more open ground when physically able. • Presses SOS only when the situation genuinely requires it. • Waits for an acknowledgement or sign that rescue is coming. • Consciously conserves battery once in a low-coverage stretch.	FEELS • Mild unease entering a known dead-zone stretch. • Isolation once every signal indicator disappears. • Fear and urgency at the actual moment of injury or distress. • Uncertainty in the silence right after pressing SOS. • Relief once an SOS acknowledgement is actually confirmed. • Renewed confidence once location has been successfully transmitted.
PAINS • No cellular network across most of the trekking route. • Difficult terrain — valleys, dense canopy, ridgelines that block signal further. • Uncertainty about whether a reported location is actually accurate. • GPS fix can take noticeable time to acquire, right when time matters most. • Battery is limited and already shared across phone, camera, and torch. • No nearby communication device in genuinely isolated stretches. • No way to know whether an SOS actually reached anyone. • Total dependence on however long rescue response takes once alerted.	GAINS / NEEDS • A simple, fast way to activate an SOS under stress. • GPS location transmitted automatically as part of the alert. • Long-range LoRa communication that doesn't depend on cellular towers. • A clear acknowledgement that the SOS actually left the device. • Low-power operation that doesn't compete with other essential gear. • A portable design that adds minimal weight or bulk. • Minimal technical interaction — no networking knowledge required. • Passive benefit from any nearby TerraLink user's device forwarding for them. • Confidence that rescue teams have a real chance of locating them.

KEY USER INSIGHT:

1. The **user does not primarily need another communication gadget**; they need confidence that an SOS carrying their location can reach someone when cellular infrastructure simply isn't there.
2. Silence after pressing SOS is as distressing as the emergency itself — **a clear acknowledgement matters** as much as the transmission.
3. Reliability depends on whether another TerraLink user happens to be nearby — this is a real, honest constraint the design must communicate rather than obscure

USER PERSONA:



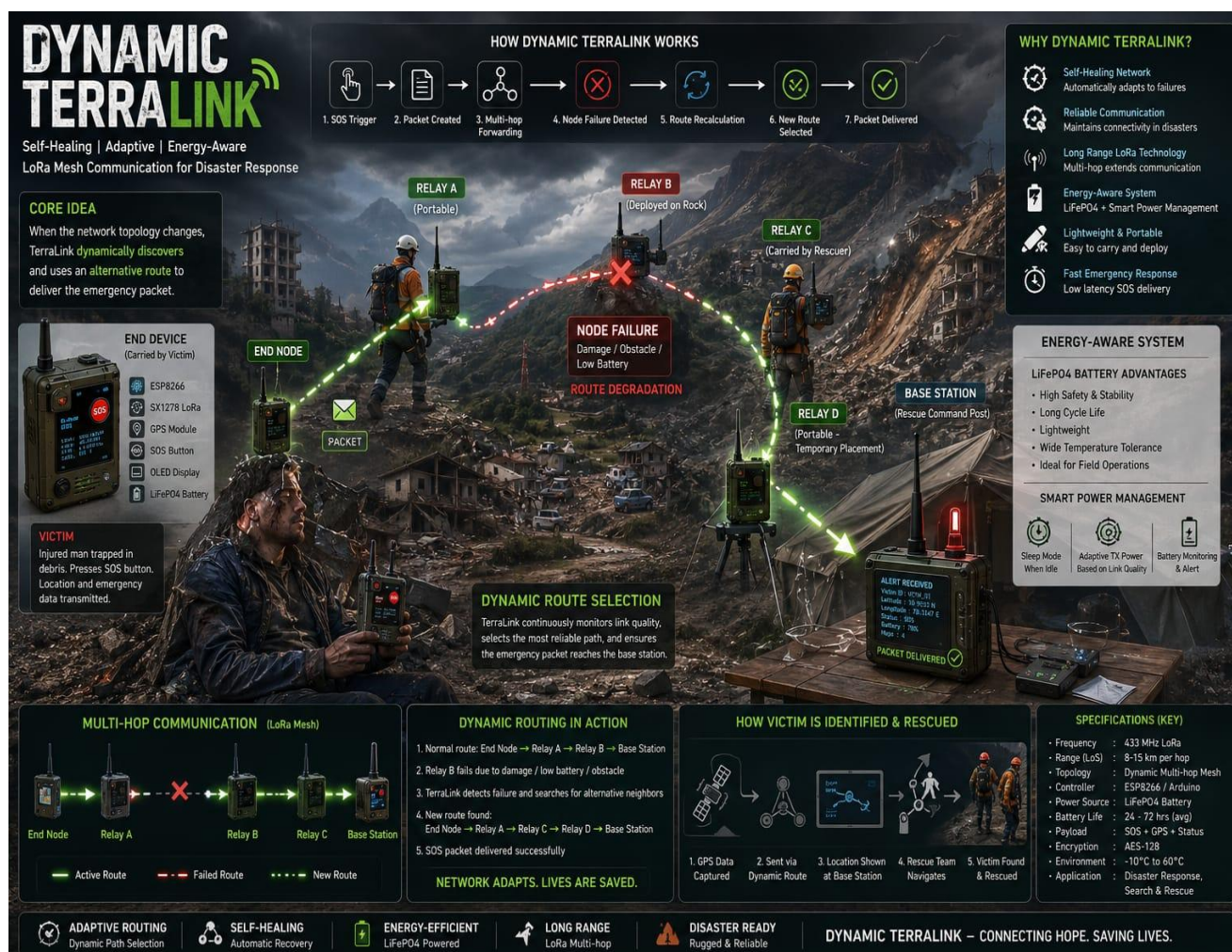
User Persona – Uma

Uma, 28 | Trekker / Forest Explorer

- **Goal:** Get emergency help quickly in remote areas.
- **Needs:** SOS, GPS location sharing, and reliable communication.
- **Pain Points:** No cellular network, difficult terrain, and limited battery.
- **Behaviour:** Uses GPS/offline maps and conserves battery.
- **Expectation:** A simple device that sends **SOS + location without cellular coverage.**
- **Core Need:**
"Reliable emergency communication in network-dead zones"

PROPOSED SOLUTION:

- **Dynamic TerraLink** is a LoRa-based, multi-hop emergency communication system designed to maintain connectivity when conventional communication infrastructure is unavailable.
- TerraLink **dynamically adapts its communication route** when nodes or links become unavailable, enabling continued emergency communication without relying on cellular infrastructure.



CORE SOLUTION :

- **SOS + GPS:** User triggers an SOS and sends their location.
- **LoRa Communication:** Provides long-range, low-power wireless communication.
- **Multi-Hop Relays:** Emergency packets travel through multiple relay nodes.
- **Dynamic Routing:** Nodes discover and select suitable communication paths.
- **Route Monitoring:** The network monitors node and link conditions.
- **Self-Healing:** If a relay or link fails, an alternative route is discovered and selected.
- **Emergency Priority:** SOS packets are given priority during network congestion.
- **Reliable Delivery:** The emergency message and location are delivered to the rescue/base station.

FUNCTIONAL REQUIREMENTS:

1.SOS Generation – Trigger SOS and create emergency packet.

2.GPS Acquisition – Capture and transmit location.

3.LoRa Communication – Send/receive emergency data.

4. Multi-Hop Forwarding – Forward packets through relays.

5. Neighbour Discovery – Identify available relay nodes.

6. Dynamic Route Discovery – Find alternative routes.

7.Route Selection – Select a suitable/reliable path.

8.Failure Detection – Detect failed/degraded links or nodes.

9.Route Recovery – Switch to an alternative route.

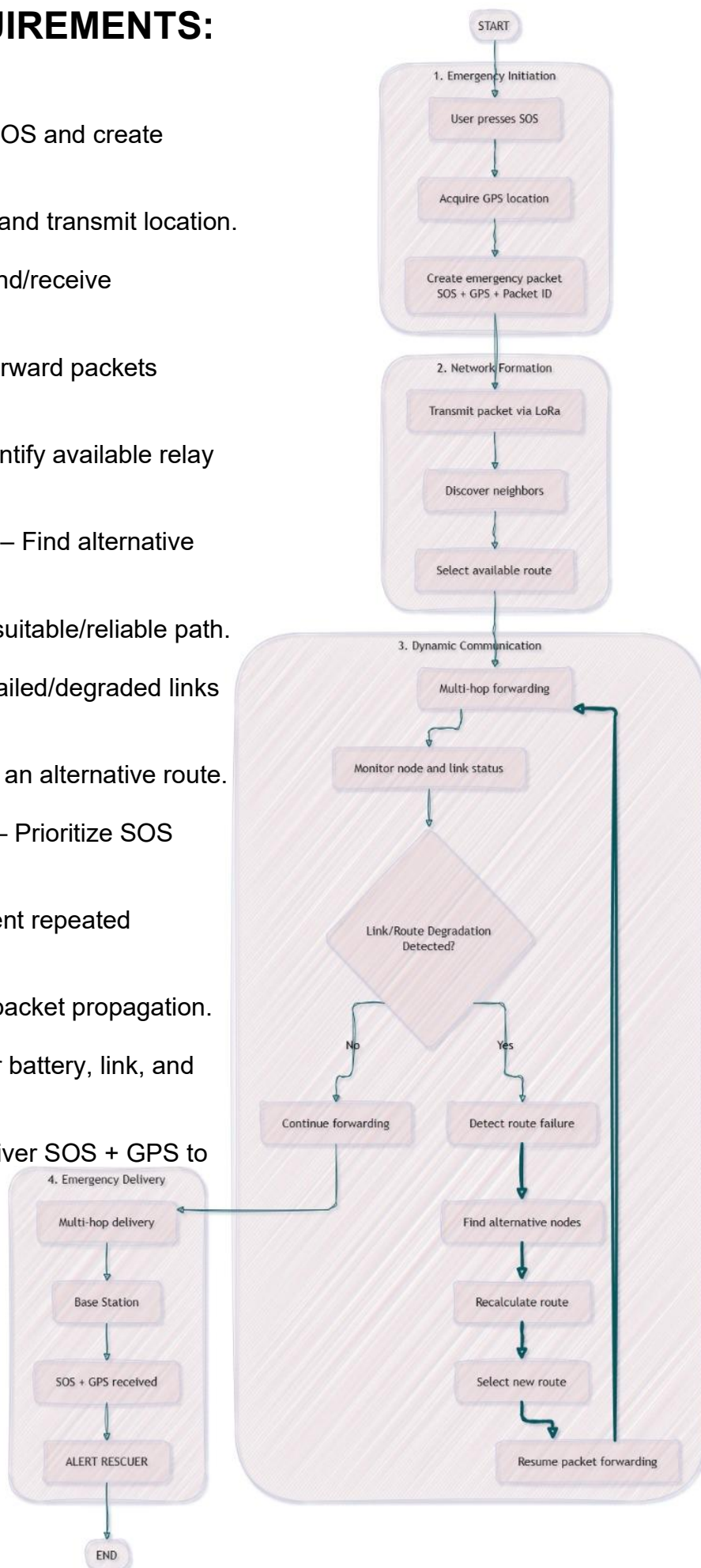
10.Emergency Prioritization – Prioritize SOS packets.

11.Duplicate Filtering – Prevent repeated forwarding.

12.TTL / Hop Control – Limit packet propagation.

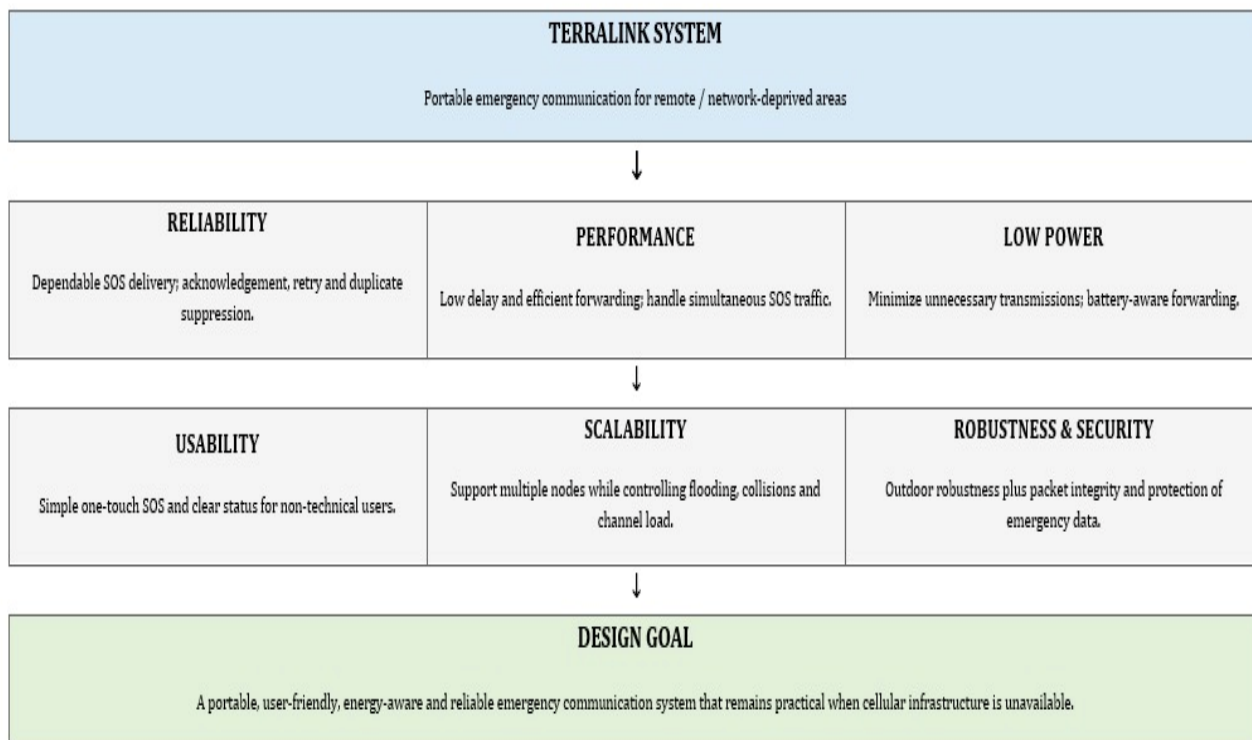
13.Node Monitoring – Monitor battery, link, and availability.

14.Emergency Delivery – Deliver SOS + GPS to the base station.



TERRALINK – NON-FUNCTIONAL REQUIREMENTS

Phase-2 Requirement Flowchart | Emergency LoRa Communication System

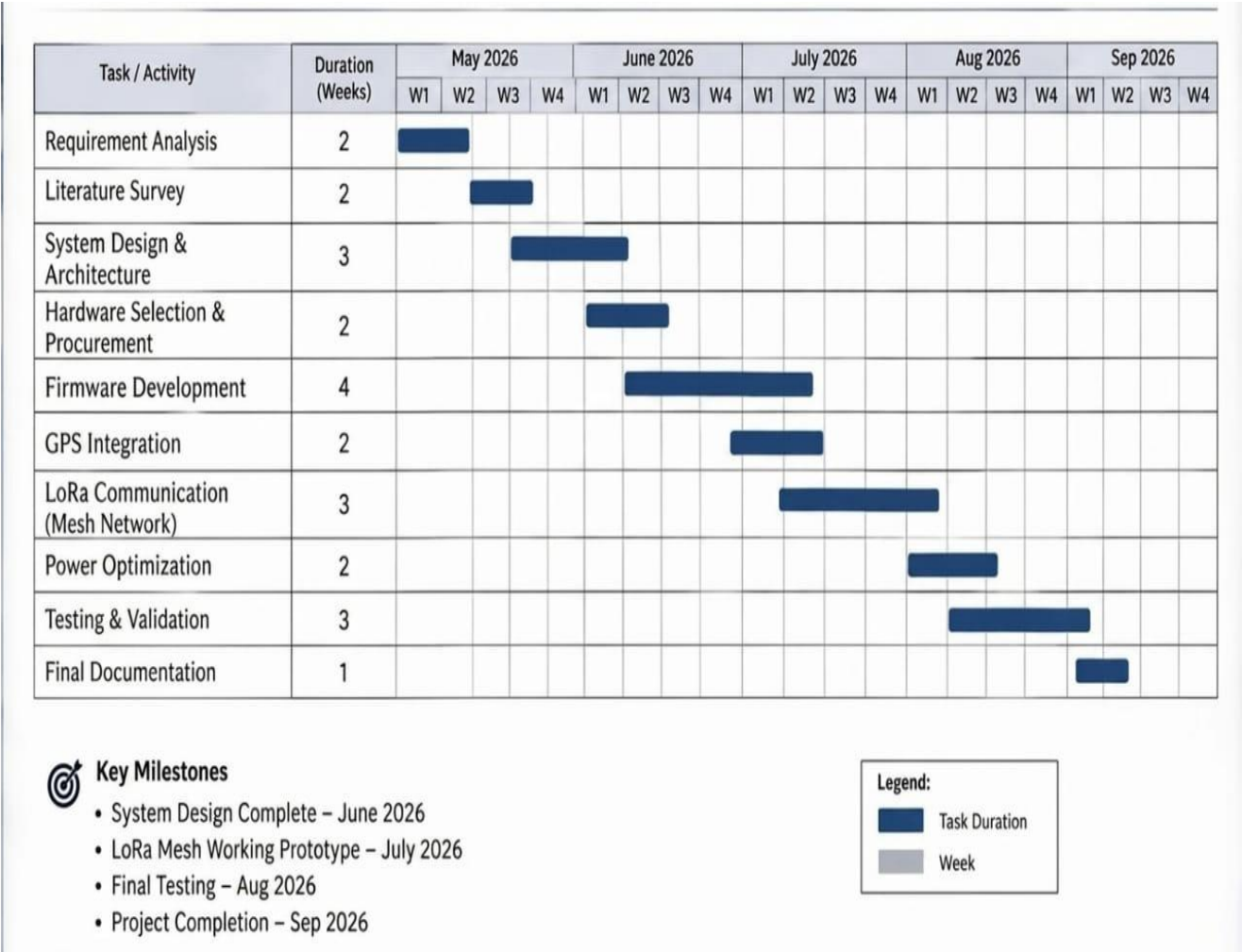


Functional requirements = WHAT the system does | Non-functional requirements = HOW WELL it should perform

Non-functional requirements define how effectively and reliably TerraLink should operate, rather than what functions it performs.

- ✓ **Reliability:** The system should deliver SOS messages reliably through available LoRa relay nodes.
- ✓ **Performance:** Emergency messages should be transmitted with minimum possible delay.
- ✓ **Range:** The system should support long-distance communication through multi-hop relay nodes.
- ✓ **Low Power Consumption:** Devices should minimize battery usage using low-power modes and controlled forwarding.
- ✓ **Scalability:** The network should support multiple TerraLink devices and relay nodes.

GANTT CHART:



This Gantt chart shows the planned timeline and sequence of activities for the TerraLink project from May to September 2026. The project begins with requirement analysis and a literature survey, followed by system design, hardware selection, and firmware development. GPS integration and LoRa mesh communication are then implemented and tested. The project continues with power optimization, system testing, and validation. Finally, the project documentation is completed.

The major milestones include system design completion, development of a working LoRa mesh prototype, final testing, and project completion. This schedule helps the team manage time, track progress, and complete each stage of the project within the planned period.

RISK ANALYSIS:



HIGH-PRIORITY RISKS:

- Node/relay failure** → self-healing multi-hop routing
- Battery depletion** → power monitoring + low-power design
- Message loss** → acknowledgement + retransmission
- GPS failure** → last valid location + GPS status
- Limited range** → strategically placed relay nodes

Overall: The major risks in TerraLink are related to **power, communication reliability, node failure, and location accuracy**. The system addresses these through **multi-hop networking, redundancy, acknowledgements, power management, and GPS validation**.